RESEARCH IN SCIENCE & TECHNOLOGICAL EDUCATION

01/03

NEW ARTICLE

Structural quality and scientific credibility in students' arguments: a comparative analysis of supporting and opposing perspectives

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DOI **doi.org/10.1080/02635143.2026.2738944**

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WHAT DID THE RESEARCHERS INVESTIGATE?

- Middle school students often find it difficult to write arguments that are both structurally organised and scientifically credible.
- The study analysed supporting and opposing written arguments from 109 eighth-grade students.
- The arguments addressed three curriculum-aligned cases: nutrient cycling and plant vitality, thermal conductivity, and lightning formation.

Bezci (2026) · Research in Science & Technological Education · doi.org/10.1080/02635143.2026.2738944

WHAT DID THEY FIND?

- Students relied mainly on referential evidence and explicit scientific theory explanations.
- Supporting arguments were generally stronger than opposing ones, which more often remained incomplete, weakly justified or low in credibility.
- The study suggests students may need greater instructional attention to justify claims, evaluate evidence and respond to alternative positions.

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